

GIS for Renewable Energy Planning

Three spatial analyses in ArcGIS Pro (Spatial Analyst)

Carlo Piccillo, Berlin, 2026

About this portfolio

This portfolio presents three spatial analyses for renewable energy projects: connecting a wind farm to the grid, comparing the visual impact of two wind farm sites, and estimating solar irradiation on roof surfaces. All three study areas lie in North Rhine-Westphalia. Each analysis follows the same structure: a two-page description (question, data and source, method, result, analytical limits), followed by the map.

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Background

The analyses are based on exercises from the “GIS and Geodata Expert” programme at GIS-Akademie Berlin (2026), which set the tasks, parameters and scenario data. I carried out the cable route and wind farm analyses independently; the solar analysis was demonstrated in a guided session and repeated on my own. Data checks, maps and assessments are my own work. Each description closes with the analytical limits, where the course parameters depart from planning practice.

Repository

The portfolio is also on GitHub, with one folder per analysis:

<https://github.com/cpiccillo/renewables-gis-portfolio>

Least-cost cable route for a wind farm grid connection

Gronau-Epe, North Rhine-Westphalia - Cost-distance analysis in ArcGIS Pro (Spatial Analyst)

Carlo Piccillo - Course exercise T  4 "Kabeltrasse", "GIS and Geodata Expert" programme, GIS-Akademie Berlin, 2026

1 Question

A planned wind farm near Gronau-Epe has to be connected to one of three transformers on the nearest high-voltage line. Where does the cheapest cable route run, what does it cost, and which cadastral parcels does it cross, how many, and with what total area? Because the choice of connection point matters as much as the route itself, all three points were compared, not only the cheapest.

2 Data and source

Dataset	Content	Source
DTM	Digital terrain model, 1 m cells	Geobasis NRW, dl-de/zero-2-0
ALKIS	Buildings, roads, paths, watercourses, standing water, woodland, scrub, bog; 27,584 parcels	Geobasis NRW, dl-de/zero-2-0
Scenario	Wind farm transformer, 3 connection points, 19 contaminated sites	GIS-Akademie Berlin (exercise data)

The study area is the DTM extent, 4.6×4.5 km at the Dutch border. The DTM came in a custom projection equivalent to EPSG:4647 (zone number in the easting), the vector data in EPSG:25832; its cell size of 0.999889 m shows it had already been resampled before delivery. I reprojected it once to EPSG:25832 with bilinear resampling to the 2 m cells the task allows, on an even-metre grid, and used it as extent, snap raster and mask for every later step.

3 Method

The cost surface gives a cost in euros per metre and is additive:

- Slope (Surface Parameters, 3×3 window, degrees), reclassified: $0-3^\circ \rightarrow 3$, $3-7^\circ \rightarrow 4$, $7-15^\circ \rightarrow 7$, $15-30^\circ \rightarrow 15$, $> 30^\circ \rightarrow 50$.
- Surcharges: woodland and scrub +100, paths +500, roads +1,500, contaminated sites +2,000, watercourses +2,500.
- Barriers: buildings, standing water and bog, set to NoData (no route possible).

The layers were rasterised in one batch and combined in the Raster Calculator, with `Con(IsNull())` so that "outside a polygon" counts as zero instead of spreading NoData. Distance Accumulation from the wind farm (geodesic) and Optimal Path As Line with path type "each cell" traced one route to every connection point, each with its path cost. Parcels crossed by the cheapest route were selected by intersection and summed with their official ALKIS area.

Checks: all points lie on valid cost cells and all rasters share one grid; a tool warning about zero costs (010779) was checked against the cost surface (minimum 3 €/m) and had no effect.

4 Result

Connection point	Route length	Model cost	Mean cost	Straight line
North	3.01 km	�61,942	�20.6/m	2.6 km
South	3.24 km	�83,450	�25.8/m	2.2 km
East	4.34 km	�91,603	�21.1/m	2.8 km

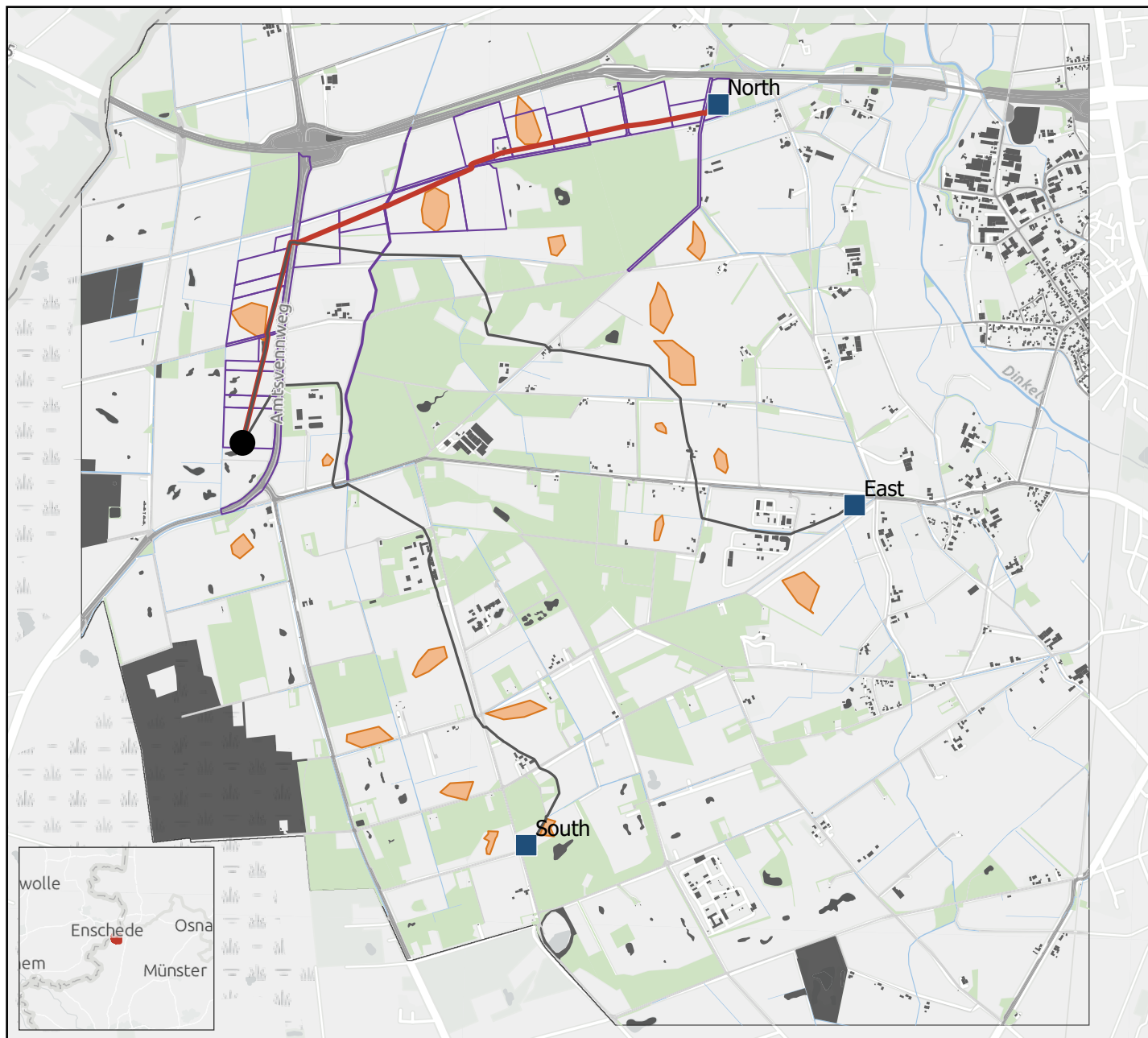
The cheapest route leads to the northern connection point: 3.0 km for  61,942. It crosses 28 parcels with an official area of 683,853 m² (68.4 ha); the geometric area is 91 m² larger (0.013 %), so the official value is the one reported.

Two findings stand out. The nearest point is not the cheapest: South is closest in a straight line, but its route costs 35 % more than North. And cost

is driven by crossings, not length: 3 km at the base rate would cost about €9,000, so roughly 85 % of the total comes from a few crossings. Slope hardly matters: 89 % of the area is below 3°, under 0.7 % steeper than 15°.

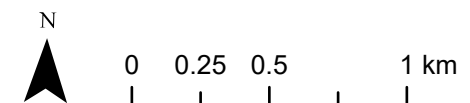
5 Analytical limits

- Per-metre costs cannot tell crossing from following. Crossing a road would in practice be one horizontal drilling at a one-off cost; the model charges it cell by cell.
- The route looks realistic for the wrong reason. For almost a kilometre it runs alongside the Amtsvennweg, but just outside the road parcel, on private land, because the parcel itself costs €1,500/m. Real cables go inside the public road parcel, in the verge, precisely because it has a single owner.
- Parcels are an output, not a cost. The task sums whole parcel areas (68 ha) for a cable a few metres wide; what matters for land rights (owners, metres per parcel, easement strip) is not in the model.
- The connection point is not the planner's choice alone. Under § 8 EEG the default is the suitable point with the shortest straight-line distance (here South), unless another is technically and economically more favourable, and the grid operator may assign a different one. The comparison informs that decision; it does not make it.
- Slope at 2 m measures micro-relief (ditch banks, embankments, DTM noise under vegetation), not the gradient relevant for trenching.
- Not modelled: protected areas, water protection zones, existing utilities, archaeology, soil, grid capacity. Costs are model values from the exercise parameters, not engineering estimates.



Point	km	€	€/m
North	3.01	61,942	20.6
East	4.34	91,603	21.1
South	3.24	83,450	25.8

- Wind farm
- Grid connection points
- Least-cost route - North (€61,942, 3.0 km)
- Alternative - East (€91,603, 4.3 km)
- Alternative - South (€83,450, 3.2 km)
- Parcels crossed (28)
- Contaminated sites - €2,000/m
- No-go: buildings, standing water, bog
- Roads - €1,500/m
- Paths - €500/m
- Watercourses - €2,500/m
- Woodland and scrub - €100/m
- Study area



Least-cost cable route for a wind farm grid connection

Cost-distance analysis on a 2 m grid - Gronau-Epe, North Rhine-Westphalia

Data: ALKIS and DTM © Land NRW / Geobasis NRW, dl-de/zero-2-0. Wind farm, connection points and contaminated sites: exercise scenario data (GIS-Akademie Berlin). Costs are model values based on the exercise parameters, not engineering estimates. CRS: ETRS89 / UTM zone 32N (EPSG:25832). Map: Carlo Piccillo, 2026. Sources: Esri, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community

Visual impact of two wind farm sites on recreation areas

Gronau, North Rhine-Westphalia - Visibility analysis in ArcGIS Pro (Spatial Analyst)

Carlo Piccillo - Course exercise T  3 "Windparkplanung", "GIS and Geodata Expert" programme, GIS-Akademie Berlin, 2026

1 Question

Two sites are under discussion for a wind farm of four turbines near Gronau. Which one affects the view from sports, leisure and recreation areas less? The course sets the threshold: the turbines are 240 m tall, a view counts as an impairment once 10 % of a turbine is visible, and people see from 1.6 m above the ground. In operational terms, an area is affected where at least the top 24 m of a turbine can be seen.

2 Data and source

Dataset	Content	Source
Surface model (DOM)	Buildings and vegetation included; 1 m cells, heights in m, no vertical reference	Course data; format matches the Geobasis NRW DOM1 (inferred, no metadata supplied)
Basis-DLM, settlement land use	1,232 polygons, of which 53 sports, leisure and recreation areas (125.6 ha) were used	Geobasis NRW (object IDs "DENW..."), dl-de/zero-2-0, via course data
Wind turbines	8 turbines at two sites (4 + 4), 150 m polylines with offsets of 216 m and 1.6 m	GIS-Akademie Berlin (exercise scenario)

The study area is the extent of the surface model. It covers the municipality of Gronau (Westf.), including Epe, at the Dutch border, within a 12.1×14.4 km bounding box.

3 Method

- Threshold. The 10 % rule is built into the turbine data: an observer offset of 216 m (90 % of 240 m) and a target offset of 1.6 m. If a sightline from 1.6 m above the surface reaches 216 m above a turbine base, the top

24 m of that turbine are visible. Sightlines are symmetric, so the turbines act as observers and every cell as a potential viewer.

- Observers. Each turbine is a straight 150 m line with two vertices. All eight have the same length and orientation (NNW-SSE), plausibly the rotor diameter seen from above. The tool uses line vertices as observers, which gives eight observer points per site; the output maximum of 8 confirmed this.
- Viewshed. Geodesic Viewshed with all sightlines, frequency output, refraction 0.13 and no distance limit; turbine base heights from the surface model; a 2 m grid snapped to the model, with the model's extent. Cells that no observer sees are NoData.
- Recreation areas. Basis-DLM features with OBJART 41008 (sports, leisure and recreation areas). Cemeteries form a separate class and are excluded.
- Visible area. Zonal Statistics as Table per area, with NoData ignored: visible cells $\times 4 \text{ m}^2$ give the area with a view of the turbines. The zone field is the Basis-DLM object ID, so each value traces back to the official object.

Checks: both outputs reach the maximum of 8 observers, and $\text{AREA} = \text{COUNT} \times 4$. The viewsheds clipped to the recreation areas contain exactly the cell totals of the zonal statistics (103,918 and 169,237). A tool warning (010451) said the heights are not ellipsoidal; this amounts to a near-uniform vertical shift of the whole model, which leaves the relative geometry, and so the visibility, practically unchanged.

4 Result

Site	Area with a view	Share of 125.6 ha	Areas affected
Wind farm 1 (south-west)	41.6 ha	33 %	52 of 53

Site	Area with a view	Share of 125.6 ha	Areas affected
Wind farm 2 (north-east)	67.7 ha	54 %	53 of 53

Wind farm 1 has the smaller impact. Its turbines are visible from 26 ha (39 %) less recreation area than those of Wind farm 2.

Two findings qualify this. First, counting areas does not separate the sites: 52 of the 53 areas see at least a turbine tip of both. Second, over the whole study area the two sites are almost equally visible, from about 46 and 49 km². The difference comes from location, not from general visibility: several recreation areas close to Wind farm 2, at the northern edge of Gronau, are visible almost in full.

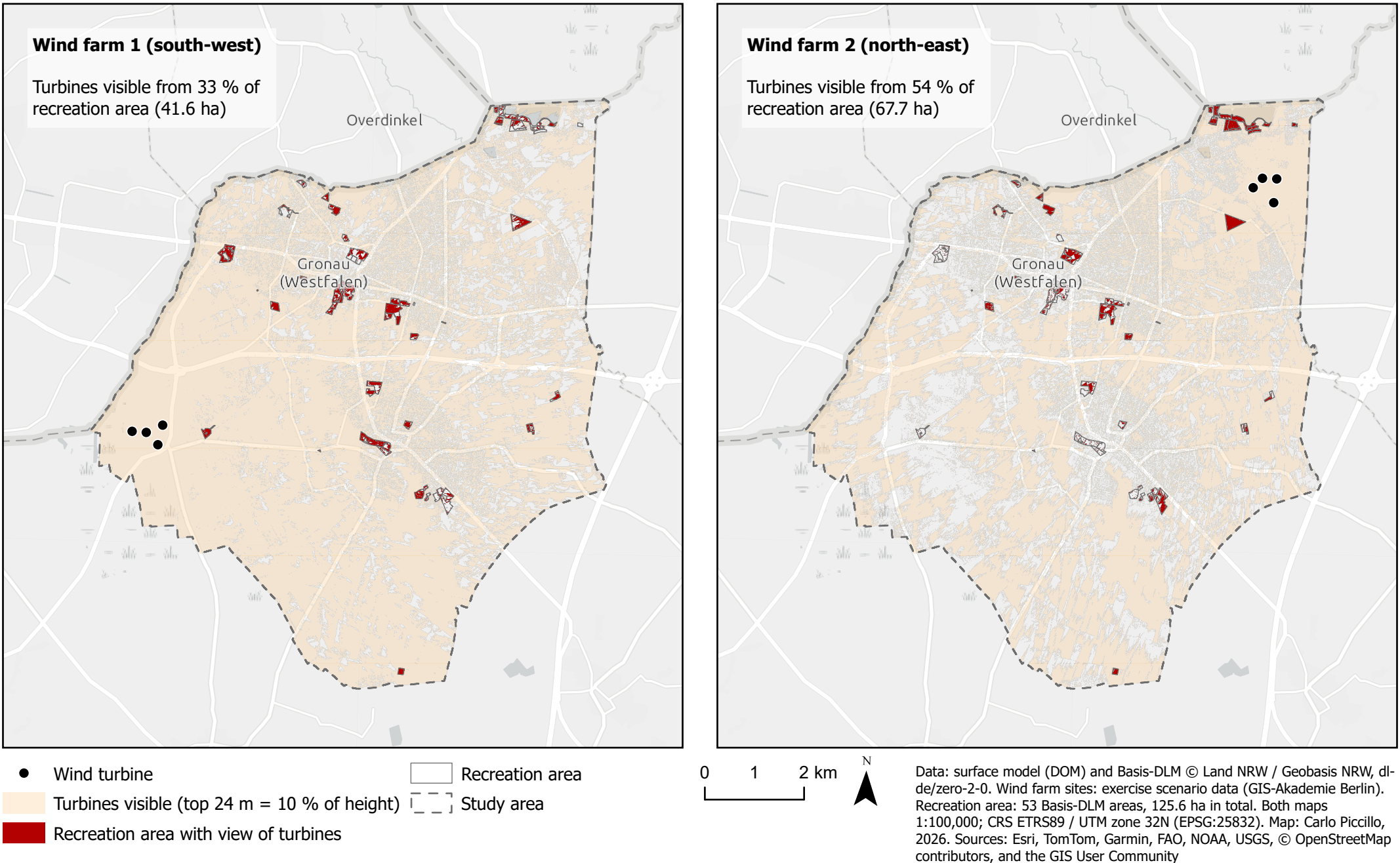
5 Analytical limits

- The perimeter decides what “less impact” means. Only recreation areas of the Basis-DLM settlement domain inside the surface model count. Woodland, open countryside and paths are not in the layer, although much of the recreation in the Münsterland happens there. The model also stops at the Dutch border, right next to Wind farm 1, so its visibility towards the Netherlands is not counted. The ranking holds within the exercise’s perimeter, not in absolute terms.
- Visible is not the same as disturbing. The threshold is binary: a 24 m blade tip 10 km away counts as much as a full turbine at 1 km. Distance, the number of turbines in view, rotor movement and night lighting are not weighted. The viewshed does count up to 8 observers, so a graded reading is possible.
- A surface model is not the ground. It includes roofs and tree crowns, so where they exist the viewers stand 1.6 m above them. In a wooded park the model therefore overstates what people on the paths can see. Resampling from 1 m to 2 m (nearest neighbour) also keeps or drops narrow obstacles, such as single trees and hedges, almost at random.
- The turbine geometry is a simplification. The two observers of each turbine sit 75 m either side of the tower, at 216 m. If the line stands for the rotor, real blade tips at that distance from the tower reach only about hub height. The rotor orientation is also fixed.

- The recreation class is mixed. It includes allotment gardens and weekend or holiday-home areas, which are closer to housing than to leisure. The area-weighted total is dominated by a few large parks: per hectare, a sports ground used daily weighs no more than a rarely used green strip.
- Not modelled: noise, shadow flicker, legal distance rules and actual visitor numbers. Visibility is one criterion of a siting decision, not the decision.

Visual impact of two wind farm sites on recreation areas

Geodesic viewshed on a 2 m surface model - Gronau, North Rhine-Westphalia



Winter solar irradiation on roof surfaces

Bochum, North Rhine-Westphalia - Solar radiation analysis in ArcGIS Pro (Spatial Analyst)

Carlo Piccillo - Guided course exercise GÜ 08 "Solarstrahlung", "GIS and Geodata Expert" programme, GIS-Akademie Berlin, 2026

1 Question

How much solar energy reaches each roof surface in the study area during the first ten days of the year, in watt-hours per square metre? The course sets the atmosphere: 40 % of the radiation arriving at the top of the atmosphere reaches the ground, and 40 % of that arrives indirectly (the course text says "by reflection"; physically it is mostly scattering). In the tool these become a transmissivity of 0.4 and a diffuse proportion of 0.4.

2 Data and source

Dataset	Content	Source
Surface model (DOM)	Buildings and vegetation included; 725 × 342 cells of 1.000 × 0.9996 m (resampled); heights 82-156 m; ETRS89 / UTM 32N	Course data; no metadata supplied
Orthophoto	RGB, same extent; used only as map background	Course data
Roof surfaces	62 polygons, geometry only; adjacent pairs indicate one polygon per roof plane	GIS-Akademie Berlin (exercise data)

The study area is the extent of the surface model, about 725 × 342 m in Bochum (51.51° N, 7.22° E). The 62 surfaces lie in its southern part; the allotment gardens to the north are not included, and the course does not say how the surfaces were chosen.

3 Method

- Settings. Extent, cell size and snap raster were set to the surface model before any raster tool. Its cells are not exactly 1 m, so "same as DOM" rather than a typed value.
- Irradiation. Raster Solar Radiation on the whole surface model, from 1 January 00:00 to 11 January 00:00 (UTC); uniform sky, diffuse proportion 0.4, transmissivity 0.4; slope and aspect derived by the tool from 3 × 3 cells. No mask: the shade of neighbouring trees and buildings has to enter the calculation. Outputs: total irradiation (kWh/m²) and direct-sun duration (h).
- Roof edges. At 1 m, the cells along a roof outline partly show walls and ground. The polygons were shrunk by 1 m (Pairwise Buffer, -1 m, no dissolve), so each mean rests on interior cells only.
- Values per surface. Zonal Statistics as Table (all statistics) on the shrunk polygons, with the original object ID as zone field; mean × 1,000 gives Wh/m². The same statistics on the original polygons serve as a comparison. Both were written to a copy of the roof layer with Join Field.
- Reliability. A surface without interior cells gets no value; one with fewer than 10 interior cells (10 m²) is flagged as low confidence.

Checks: the longest direct-sun duration, 77.8 h, is just below the roughly 80 h of daylight at 51.5° N over these ten days, so sun geometry and horizon behave as expected. AREA matches COUNT × 1 m². Warning 010566 (some zones not rasterized) matches the five shrunk polygons that contain no cell centre; a sixth vanished in the buffer. An intermediate copy of the roof layer had renumbered its object IDs, which shifted values between roofs from row 9 on; the join was rebuilt on a copy that keeps the IDs and checked against the zonal tables.

4 Result

Roof surfaces	Minimum	Median	Maximum
56 of 62 with a value	273 Wh/m ²	504 Wh/m ²	1,591 Wh/m ²

South-facing planes receive the most: up to 1,591 Wh/m², nearly six times the least favoured surface. The distribution is bimodal: 34 of the 56 surfaces receive less than about 600 Wh/m², 7 more than about 1,430, and few lie in between, so the median describes a typical surface better than the mean (729). The map shows the pattern: on buildings with an east-west ridge the south plane falls in the upper classes and the north plane in the lower ones, while planes facing east or west sit in between. With the sun at most about 16° above the horizon, the orientation of a plane decides more than where the building stands.

Edge cells matter for single surfaces, not for the overall picture. Against the original polygons, the shrunk ones change a surface's mean by a median of 6 % (mean 9 %), in both directions (median signed difference -0.1 %). Differences of 15-23 % occur on mid-sized planes too, and up to 78 % on the smallest. Six small roof sections (5.6-8.2 m²) have no value, and 8 surfaces rest on fewer than 10 cells, two on a single cell.

5 Analytical limits

- The absolute values are far too low; the ranking is the result. DWD's long-term January mean for Germany is 23 kWh/m² on a horizontal surface (1981-2020), roughly 5-7 kWh/m² for these ten days. The model gives at most 1.9 kWh/m² for any cell and 1.6 for the best roof plane, which faces south and in winter should receive more than a horizontal surface. In Esri's model, transmissivity refers to the vertical path and is raised to the relative path length, about 3.6 at a 16° sun: $0.4^{3.6} \approx 0.04$, so little direct radiation gets through. The course script gives 0.4 as a typical value for Germany and points to the Global Solar Atlas for more precise ones.
- Ten January days are not a year. With a low sun, orientation and shade dominate; over a year, flat and east- or west-facing roofs catch up. The values say nothing about annual yield.
- Duration is geometry, not weather. The 77.8 h are hours in which the sun stands above the local horizon; clouds enter only through the constant transmissivity.

- The surface model sets the horizon. Obstacles beyond its edge are missing, so planes near the southern edge may be overestimated: at this sun height a 10 m obstacle casts a shadow of about 35 m. The orthophoto shows summer foliage; if the model was captured at the same time, deciduous trees shade more in the model than in January.
- Orientation was read from the map, not measured. Aspect per plane was not extracted, so the explanation by orientation rests on the visual pattern.
- Not modelled: module efficiency, temperature, inverter losses, usable area (setbacks, chimneys, dormers) and roof statics. Irradiation is one input to a solar-potential assessment, not the yield.

Winter solar irradiation on roof surfaces

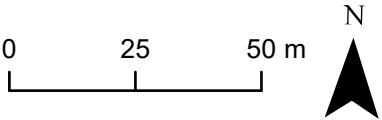
Mean solar irradiation per roof surface, 1-10 January - Bochum, North Rhine-Westphalia



Irradiation, 1-10 January (Wh/m²)

- 270 - 450
- 450 - 550
- 550 - 750

- 750 - 1200
- 1200 - 1591
- Low confidence (< 10 m² interior)
- No value (excluded)



GIS-Akademie Berlin course exercise (GÜ 08
Solarstrahlung)
Data: course dataset - 1 m DSM, orthophoto, roof
surfaces
Method: Raster Solar Radiation (uniform sky, diffuse
0.4, transmissivity 0.4); mean per roof surface
buffered -1 m
ETRS89 / UTM 32N - Map: Carlo Piccillo, October 2026